

Insilico Codon Bias Correction for Transgenic Biological Protein Sequences for Vaccine Production

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Abstract—Codon optimization is primarily used in enhancing the levels of protein expression in the host species. Each species has its own codon usage bias, which represents the codons abundance frequency in that species. Using the host usage profile contributes to personalize the synthesis of the DNA vaccines that can achieve highly active vectors the host cells. For optimizing protein expression levels in a particular host, the genetic code sequence needs correction of codon frequency bias to match the expression of host codon landscape rather than the donating organism profile. In this work, we have applied two approaches for optimizing codon usage in protein-coding sequences. The first approach adopts a substitution-based method to replace less frequent codons with differentially higher frequency codons at the specific codon usage bias tables. The second approach finds and replaces the maximal exact protein matches between the unoptimized sequence and the host proteome. We evaluated our work by optimizing the Avian Influenza H1N1 virus's HA gene to maximize the protein expression before synthesizing the DNA vaccine. Our method produced optimized sequences with higher GC content by 17%, which is similar to eukaryotic sequence profiles than the viral usage profiles, allowing for better expression in avian host cells.

I. INTRODUCTION

The central dogma of molecular biology governs a set of rules that maintain protein expression in living cells. DNA coding gene information is transcribed into mRNA, which is an intermediate molecule that is further translated into protein. This occurs with the help of a key amino acid carrier molecule called tRNA. These tRNA molecules form a specific link between nucleotide triplets in the mRNA and the amino acid encoded by each triplet in the growing peptide chain. Ribosomes are known to be protein production machinery in the cells, and it can adapt to both mRNA and tRNAs molecules. Almost all known living organisms have the same codon translation table for translating amino acids. This genetic code is formed of 61 different nucleotide triplet variations called codons. These codons are encoding 20 standard amino acids. Hence, several amino acids will have more than one representative codon, where multiple triplets encode 18 out of the 20 amino acids called synonymous codons [1]. Synonymous mutations are considered not to affect protein function and denoted as “silent mutation”. However, the distribution of synonymous codons in the genome didn't always follow

a random distribution. Comparative sequence analysis for different organisms showed that each organism has a preferred set of codons over the others and tends to use them more frequently. This unique usage pattern in each organism is known as codon frequency bias [2].

II. RELATED WORK

Codon bias has been subject to an immense number of studies in the past years. It has been found to affect associated protein expression levels and protein 3D folding [3]. Other studies correlated the codon bias with differential protein expression [4]. Recent advances in the field of de novo DNA synthesis with continuously falling costs have allowed custom genes synthesis with their codon frequency optimized for certain host expressions. The possible combinations for synonymous codon alterations in one gene are significant; for a 300 amino acid long protein, there are more than 10,100 different possible variations of coding sequence alterations that can give the same protein [5]. Hence, various computational methods are being developed to manipulate synthetic coding sequences to optimize their codon usage for enhanced protein expression levels in specific other host organisms [6].

Several codon optimization commercial services are available and provided by DNA synthesis companies. Most of these methods use different algorithms to maximize the Codon Adaptation Index (CAI) of the optimized sequence to mimic that of the expression host, considering some essential sequence properties. Such features may include proper GC-content, evading sequence repeats, avoiding specific functional motifs such as restriction sites of ribonucleases (RNase), and finally improving the output mRNA secondary structures [7]. In many cases, the impact of codon optimization using such methods can be as successful as leading to more than 1000 fold increase in the expression level, only by optimizing the coding sequence [7]. However, in certain other cases, the expression is not affected or even compromised by the currently available algorithms [7], [8]. Thus, there is ample room for improvement in the currently available algorithms. For instance, in many cases, the optimized synonymous mutations can alter protein folding due to modified codon frequency distribution. Other methods of codon optimization for synthetic DNA sequences have been reported. An algorithm called “codon harmonization”

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adjusts the used codons in the expressing host intending to maintain the original codon landscape of the gene in the donating host [9]. Hence, such an algorithm should preserve as many rare codon settings as the original sequence, which proved successful, improving the heterologous expression of specific proteins in *E. coli* [9], [10]. Another approach for optimizing transgenic sequences in the model bacterial host is using statistical methods to identify several important amino acids and their specific employed synonymous codons associated with differentially higher expression profiles. These critical codons are not necessarily the most frequent codons in the host organism highly expressed genes but were related to tRNA ability to bind on the ribosome [11].

III. MATERIALS AND METHODS

We have applied two approaches for codon optimization in the development of the software tools. The first approach is optimizing by direct codon synonymous substitution. We use the species-personalized codon usage bias table to replace the low-frequent codons with the high-frequent ones as illustrated in figure 1. The second approach is by finding maximal exact protein matches between the host and the virus, then replacing the virus's DNA subsequence with the host's DNA match. We developed two tools to apply both approaches.

A. Codon usage bias processing

We used the Codon Usage Database hosted at <http://www.kazusa.or.jp/codon> as it stores the codon usage tables of 35,800 species and three million complete protein-coding genes. Our software tool enables the user to copy-paste their host species' codon usage bias table for parsing. We then convert the codons table into a key-value data structure with the amino acid as a key and its associated codons with their frequencies of occurrence as a value. Querying this data structure will decide whether to replace a codon by its best alternative synonymous one or not.

B. Codons substitution

The application enables the user to copy-paste or uploads a Fasta DNA sequence to be optimized. We convert the DNA sequence to a protein sequence then keep it in memory for results and further assertions. By iterating over the protein sequence, we query the codon's data structure to replace the current amino acid's codon with its best alternative.

C. Optimized sequence validation

We translate both of the sequences before and after codons-optimization to their protein sequences. As the codons-optimization performs the changes only on the DNA level, the protein sequences should stay the same in the whole process. Therefore, we asserted by comparing the two protein sequences to ensure they are the same.

D. Output representation

Our tool focused on delivering a very detailed output to the user for further analysis. The final output is a three sections web page. The first section contains the original DNA sequence, the optimized DNA sequence, and the protein sequence. Part

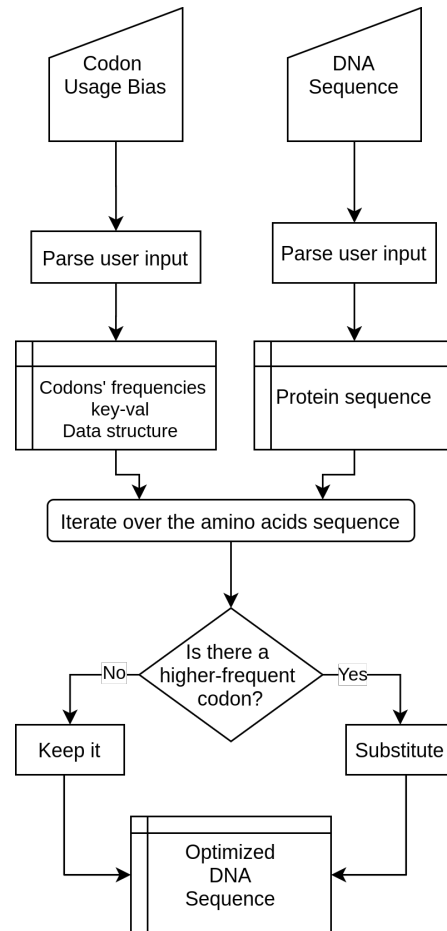


Figure 1. A flowchart showing the heuristics of the codons optimization by synonymous substitution.

two has a detailed and user-friendly table of all the changes that occurred to the original sequence. Columns of the table are serial ID, codon position, the old codon sequence, the optimized new codon sequence, the difference between the new and old codon frequencies, the amino acid corresponding to the codons, and the fractions of all the synonymous codons highlighting the selected frequency for replacement.

E. Non-model organisms support

Some organisms do not have an entry in the codon usage bias database. To address this problem, we have developed another approach that does not require the codon usage bias table for codon optimization. The proposed method relies on the protein maximal similarity method, where the usage profiles are not inferred from codon frequency distribution over the host genome. Instead, it looks after the most extended protein polypeptide chain similarity then uses the corresponding DNA encoding for it as an optimization input for our target protein. In other words, we cut the target protein into smaller segments and search the host proteome for similar subsequences. Maximal exact matches are retrieved and translated back into their corresponding DNA sequences. These DNA segments are used

Changes Table (Click to expand)						
Optimized Codons						
#	pos	Old	New	diff	amino	Fractions
1	1	GTT	GTG	0.24	V	- GTT : 0.21 - GTC : 0.22 - GTA : 0.12 - GTG : 0.45
2	4	AAA	AAG	0.12	K	- AAA : 0.44 - AAG : 0.56
3	7	AGT	AGC	0.12	S	- TCT : 0.18 - TCC : 0.2 - TCA : 0.15 - TCG : 0.07 - AGT : 0.14 - AGC : 0.26
4	16	ATT	ATC	0.11	I	- ATT : 0.35 - ATC : 0.46 - ATA : 0.18
5	22	ATT	ATC	0.11	I	- ATT : 0.35 - ATC : 0.46 - ATA : 0.18
6	25	GGT	GGC	0.13	G	- GGT : 0.18 - GGC : 0.31 - GGA : 0.27 - GGG : 0.25

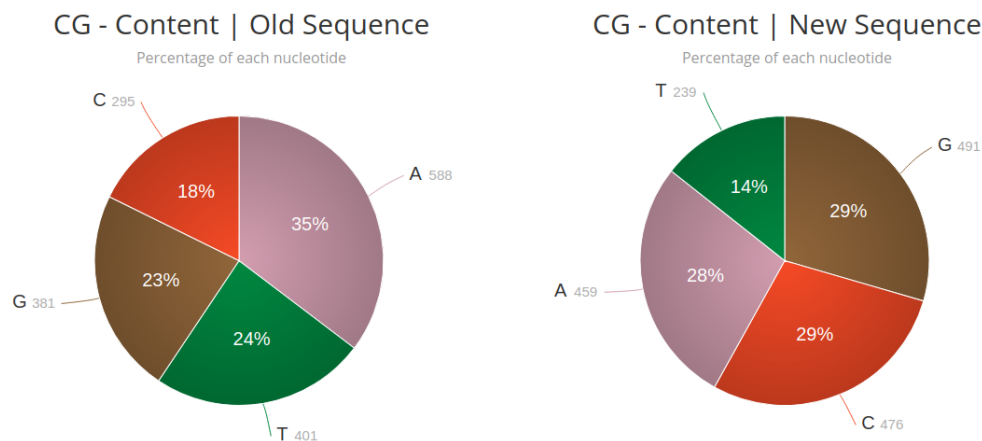


Figure 2. A screenshot showing the detailed UI of the results on the web application. The table represents all the codon changes performed in the optimization, and the pie chart represents the GC-content before and after optimization.

for rebuilding the transgenic DNA gene to produce the same protein but using building components that are more similar to host cells.

F. DNA-Vaccine experiment

We attempted to optimize the Influenza A virus gene H5N1 that targets the chickens. There are multiple copies of that gene depending on the environment, so we selected the Egyptian sequenced copy of the virus gene with the accession number (GenBank: HQ198268.1). We used the codon usage bias of the host organism Gallus gallus and then performed the codon optimization using our tool.

IV. RESULTS AND DISCUSSION

Figure 2 shows the friendly user interface of our tool with detailed results. The result of running the virus optimization experiment showed the production of codons usage profile that is more realistic and similar to the host usage profile. The GC content of the optimized sequence has increased by 17%, which will consequently make a better gene expression due to approaching the known GC-content levels in the eukaryotes.

V. CONCLUSION AND FUTURE DIRECTIONS

We proposed two different methods for codon optimization which can be impactful in core biotechnological applications like the DNA vaccination. Our tool has a user friendly interface to be easily used by biologists. We plan for further evaluation for the optimized DNA vaccine by invitro experiments.

VI. SOFTWARE AVAILABILITY

Both tools are open-source and publicly available on Github. The synonymous substitution-based codon optimizer is available at <https://github.com/mr-eyes/web-codonsOptimizer> and <https://mr-eyes.com/nu/cop>. The MEM-based codon optimizer is available at <https://github.com/mr-eyes/MEM-codons-optimizer>.

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